



Out into the Open: Occurrence of Extended-spectrum β -lactamase-producing and Multidrug-resistant *Escherichia coli* in Mariculture and Recreational Coastal Waters

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Abstract The Tricycle Protocol is a One Health initiative that involves monitoring extended-spectrum β -lactamase-producing *Escherichia coli* (ESBL-Ec), an opportunistic intestinal and extraintestinal pathogen that is recognized as critically important for the development of new antimicrobial agents and as a key indicator of water, food, and environmental contamination. This study aimed to detect and characterize, phenotypically and genotypically, the presence of ESBL-Ec in coastal waters collected from three urban marine ecosystems and mussels from a mariculture area in the metropolitan region of Rio de Janeiro

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City, Brazil. Eighteen samples from each beach and from mussels were collected between June 2022 and December 2023. A total of 299 *E. coli* strains were identified over the study period, of which 33 (11%) were initially identified as potential ESBL producers after selective pressure cultivation. Antimicrobial susceptibility analysis revealed a high frequency of resistance to ampicillin and third-generation cephalosporins, exceeding 80%. Eighteen (54.4%) of the 33 *E. coli* strains were classified as multidrug resistant (MDR). Genotypic confirmation revealed at least one ESBL gene (*bla*_{CTX-M-1,2}, *bla*_{CTX-M-8}, *bla*_{CTX-M-14}, *bla*_{TEM}, or *bla*_{SHV}) in 30 of the 33 strains, with *bla*_{CTX-M-1,2} being the most prevalent in samples from one estuarine beach and from the mariculture site. Additionally, the genes *qnrS*, *intI1*, and *intI2* were detected in 21.5%, 7.6%, and 1.3% of the confirmed ESBL-Ec strains, respectively. Since we could not associate the presence of MDR ESBL-Ec strains with unsuitable water classification, these findings underscore a grave threat to both human and animal health in densely inhabited coastal regions.

Highlights

- Waters and mussels samples from different sources were analysed for ESBL-Ec presence.
- No correlation between the number of isolates and any of the physical-chemical parameters, not even for ESBL-Ec.

- Antimicrobial susceptibility analysis revealed a high frequency of resistance to ampicillin and third-generation cephalosporins, exceeding 80%.
- Up to 50% of ESBL-Ec strains were classified as multidrug-resistant.
- Up to 90% of ESBL-Ec strains were positive for at least one ESBL gene, in addition to *qnrS*, *int1*, and *int2* genes.
- It was not detect ESBL-Ec in mussels.

Keywords Antimicrobial resistance · ESBL · *Escherichia coli* · Marine environment · One Health

1 Introduction

Marine environments have been increasingly shown to contain diverse and abundant reservoirs of antimicrobial-resistant (AMR) microorganisms and antimicrobial resistance genes (ARGs). Anthropogenic activities, particularly industrial wastewater and domestic discharge, play a critical role in enriching these environments and facilitating the dissemination of ARGs among water sources (K et al., 2023; Singh et al., 2022). The ocean resistome comprises more than 1,000 distinct ARGs that putatively confer resistance to 23 classes of antimicrobial agents, including those frequently used in clinical practice, such as β -lactams and aminoglycosides. Additionally, this core component of the marine resistome demonstrates a high likelihood of horizontal genetic transfer (HGT) events, with approximately one-fourth of these transfers being associated with high health risks (Xu et al., 2023).

Genes related to sulfonamide resistance, as well as those encoding different β -lactamase families, have been detected in water samples collected from different sources worldwide, from both pristine environments and urban areas. The regions included heavily populated coastal zones with wastewater discharge, aquaculture activity, maritime traffic, and oceanic environments. These findings suggest that these genes play important roles in the introduction and spread of ARGs throughout oceans (Bonanno Ferraro et al., 2024).

Among the potential ARG bacterial reservoirs in coastal zones, *Escherichia coli* occupies a particular position both as a host and as a potential disseminator of multiple resistance determinants (Costa et al., 2023; Lübcke et al., 2024). This gram-negative

rod-shaped belongs to the *Enterobacteriaceae* family and is commonly found as a commensal microorganism in the gastrointestinal tract of humans and other animals. In these organisms, it contributes to maintaining intestinal homeostasis and microbiome stability. However, *E. coli* is also an important human pathogen, with a variety of well-characterized intestinal and extraintestinal pathotypes capable of causing diverse clinical manifestations, including gastroenteritis, urinary tract infections, meningitis, bacteremia, and septicemia (Geurtsen et al., 2022).

In addition to the variety of pathotypes within the species, a wide range of antimicrobial resistance profiles have been widely described in *E. coli* strains isolated from humans, veterinary, and environmental sources, including multidrug-resistant, extensively drug-resistant, and panresistant phenotypes (Kerek et al., 2025; Nasrollahian et al., 2024; Poirel et al., 2018). In particular, resistance to β -lactams represents a major public health challenge, not only because of the considerable diversity of β -lactamase genes but also because of their association with mobile genetic elements (MGEs). Among these enzymes, extended-spectrum β -lactamases (ESBLs) constitute a rapidly evolving group characterized by their ability to hydrolyze third-generation cephalosporins and aztreonam. They are inhibited by clavulanic acid and are frequently encoded by plasmids (Zhang et al., 2025a). Given the increasing threat posed by ESBL-producing bacteria, the World Health Organization (WHO) established the Integrated Global Surveillance on ESBL-producing *E. coli* (ESBL-Ec) using a “One Health” approach (Tricycle Protocol) (World Health Organization, 2021). This initiative aims to implement an integrated and simplified cross-sectoral surveillance system for antimicrobial resistance in bacteria. The Tricycle Protocol has been successfully implemented in various countries (Appling et al., 2023; Milenkov et al., 2024; Ruppé, 2023). Researchers have learned to emphasize the importance of tracking and understanding the presence and transmission of ESBL-Ec across human, animal, and environmental sources.

This holistic perspective facilitates epidemiological surveillance of antimicrobial resistance, which has been implemented in various studies in recent years, within the One Health framework. Livestock farming is well known as a critical source of ARGs dissemination, especially for *Enterobacteriaceae* (Karabasil

et al., 2025; Wu et al., 2024), and the implications of multidrug-resistant pathogenic bacteria in the poultry industry are a major concern due to significant losses, zoonotic risks and food safety (Ahmed et al., 2025; Ali et al., 2025). Control of non-typhoidal outbreaks was shown to be more efficient and economically beneficial when multisectoral surveillance-response was integrated into public and animal health systems (Sanni et al., 2024).

The transmission pathways, as well as the molecular evidence of ESBL-Ec dissemination, were evaluated in a review study, and the conclusion emphasized the need for integrated surveillance control strategies (Cahyani et al., 2026).

To date, several studies have reported various ARGs in different bacterial species isolated from coastal and estuarine waters in Rio de Janeiro (RJ, Brazil) (Alves et al., 2024; Canellas et al., 2021; Costa et al., 2023; Montezzi et al., 2015; Paschoal et al., 2017). With respect to ESBL-Ec in the region, Costa et al. (2023) isolated and identified 55 ESBL-Ec strains collected from water sources with three different pollution gradients in Guanabara Bay, Rio de Janeiro, Brazil. The authors suggested that these ecosystems may act as neglected reservoirs of ESBL-Ec and suggested a possible association between antimicrobial-resistant *E. coli* and pollution caused by sewage in aquatic environments. Here, we evaluated the isolation, identification, and antimicrobial susceptibility profile of ESBL-Ec in an attempt to verify the effect of anthropogenic action on the dissemination and persistence of this phenotype in marine ecosystems with different types of human impacts.

2 Materials and Methods

2.1 Water Sampling and Processing

From June 2022 to December 2023, a total of 1,500 mL of seawater was collected in three sterilized bottles to obtain a composite sample from each site. Water samples were collected at a distance of approximately 5–10 m from the coast and at depths ranging from 1 to 1.5 m, between 9 a.m. and 11 a.m. local time. Sampling was conducted in three distinct marine ecosystems of Niterói (RJ, Brazil): Icaraí (I), a residential beach area in Guanabara Bay with recreational activities in a densely

populated neighborhood with intense urban activity (22°54'16.0"S 43°07'01.3"W); Piratininga (P), which is also a residential area, but is an oceanic beach near the bay entrance and under its influence, with recreational activities in an area of moderate urban activity (22°57'11.9"S 43°05'54.0"W); and Jurujuba (J), a region in Guanabara Bay with historically relevant mariculture activity in the state (Landuci et al., 2021), but currently recognized as heavily impacted by irregular sewage discharge; at this location seawater samples were collected directly from the mariculture site (22°56'04.1"S 43°06'46.6"W) (Fig. 1). The selection of all sampling locations was based on their importance to the city and proximity to *Instituto Estadual do Ambiente* (INEA) sampling points for balneability bulletins (*Instituto Estadual do Ambiente*, n.d.), except for Jurujuba, where the sampling site was located in the mariculture area. To evaluate the presence of ESBL-Ec with its potential anthropogenic action, the water quality classification at these sampling locations was obtained from INEA reports for the years 2022 and 2023 (*Instituto Estadual do Ambiente*, 2022, 2023) (Supplementary Table 1).

The flasks were placed in a refrigerated icebox and immediately transported to the Laboratory of Enteropathogens, Veterinary, Environmental, and Food Microbiology at the Fluminense Federal University (LEMA/UFF, RJ, Brazil). The water samples were processed within 2 h of collection after they reached room temperature.

A total of 100 mL of homogenized seawater from each sampling site was filtered as described above in triplicate, with modifications (Forster & Pinedo, 2015). Filtration was performed using 0.22- μ m nitrocellulose membranes for each sampling site. Each membrane was incubated at 37 °C for 18 h to 24 h in trypticase soy broth (TSB—Kasvi, Pinhais, PR, Brazil) for enrichment to promote polymicrobial growth.

All the samples were initially analysed for temperature, pH, and salinity. The microbiological quality of Icaraí and Piratininga beaches was assessed using INEA bulletins; the water from the Jurujuba mariculture sampling site was analysed according to CONAMA guideline No. 357/2005 (amended by CONAMA guideline No. 430/2011), which establishes standards for the cultivation of bivalve mollusks intended for human consumption. According to

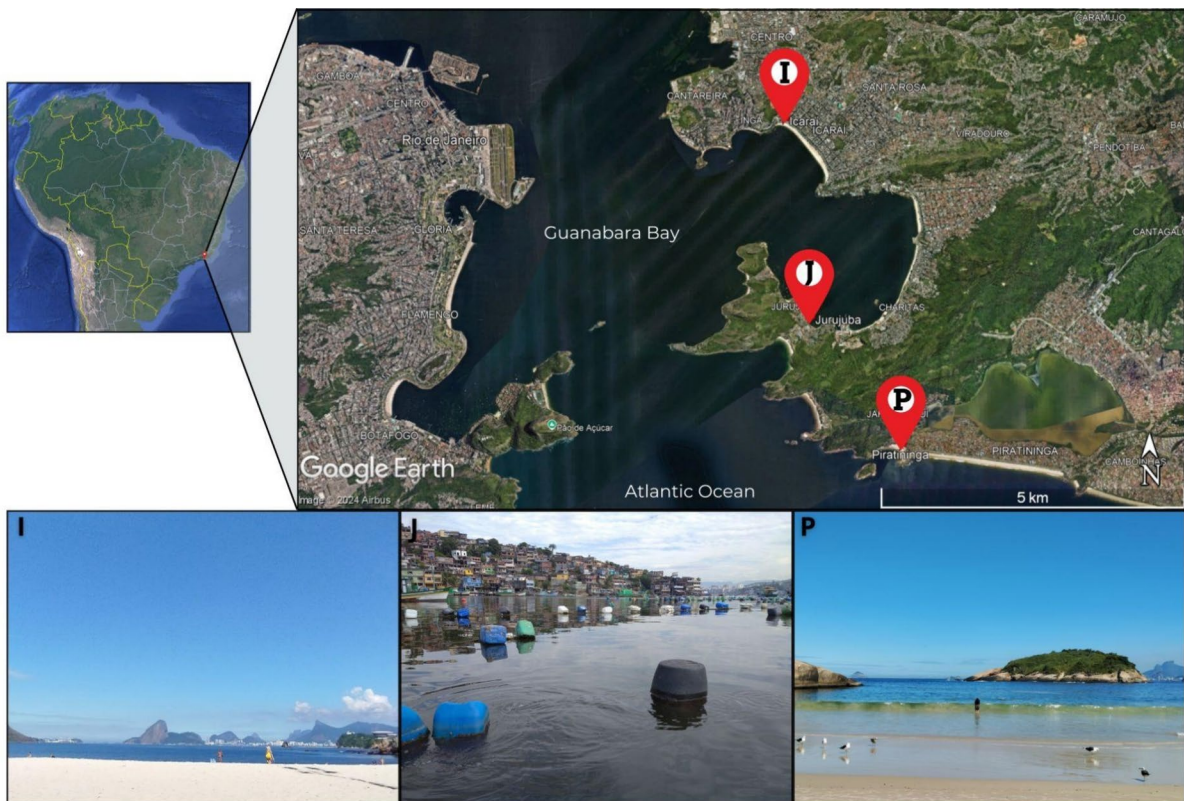


Fig. 1 Sampling sites of beach waters in Niterói city (RJ, Brazil). Scale 1 cm=5 km. I—Icarai (22°54'16.0"S 43°07'01.3"W); J—Jurujuba (22°56'04.1"S 43°06'46.6"W);

P—Piratininga (22°57'11.9"S 43°05'54.0"W). INEA sampling sites: <https://www.inea.rj.gov.br/niteroi/>

these standards, the geometric mean of thermotolerant coliforms cannot exceed 43 MPN/100 mL, and the 90th percentile cannot exceed 88 MPN/100 mL in at least 15 samples (Conselho Nacional do Meio Ambiente, 2005, 2011).

2.2 Mussel Sampling and Processing

Bivalve mollusks of the species *Perna perna*, cultivated in Jurujuba, were collected at the same site where water samples were obtained. Fifteen mussels were randomly collected, cleaned externally, stored in a cooler bag with ice, and transported to the laboratory within 2 h, where they were processed for primary isolation of *E. coli*. They were opened, macerated, and homogenized in a sterilized glass Petri dish. Next, 25 g of the mixture of the internal contents of the mussels was homogenized

in 225 mL of 0.1% (w/v) peptone water (Becton Dickinson and Company, New Jersey, USA) in an industrial blender (Walita, N°. 996924, Indústria Brasileira – São Paulo, SP, Brazil) to obtain a dilution of 10^{-1} . A serial dilution was prepared to inoculate in selective media for *E. coli* isolation in duplicate.

The analyses of coliform bacteria in mussel samples were performed using the most probable number (MPN), which was calculated and interpreted according to the MPN tables provided by the Food and Drug Administration (FDA, USA) (Human Foods Program, 2024). Following confirmation of contamination by thermotolerant coliforms, identification was performed, as described in Normative Instruction No. 161/2022, according to the parameters for fish (Ministério da Saúde, 2022).

2.3 Bacterial Isolation and Identification

To isolate *E. coli*, polymicrobial growth samples from mussels were inoculated into MacConkey agar (Merck KGaA, Millipore), eosin methylene blue agar (EMB Levine—Kasvi), cystine lactose electrolyte-deficient agar (CLED with bromothymol blue – HiMedia, Maharashtra, India), and Hektoen Enterico (HiMedia) culture media in duplicate. Following incubation at 37 °C for 24 h, colony counts were performed, and presumptive *E. coli* colonies were picked (up to five per Petri dish).

Seasonally, every three months, filtered membrane and mussel samples were also inoculated, in duplicate, into commercial CHROMagar™ ESBL (Plastlabor, Rio de Janeiro, RJ, Brazil) and CHROMagar™ KPC (Plastlabor) media, following the same processing, inoculation, and incubation procedures used for the other selective media. After incubation at 37 °C for 24 h, presumptive *E. coli* colonies (dark pink to reddish) were isolated from both chromogenic media according to the manufacturer's instructions. All presumptive *E. coli* colonies isolated in both selective and CHROMagar media were preserved in 40% glycerol (v/v) stock solution (final ratio 1:1) and stored at –20 °C.

Bacterial identification was carried out by matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS). Mass spectra were obtained using the Microflex LT System (Bruker® Daltonics), and compared with reference spectra in the spectrometer database using MALDI Biotyper 7.0 software (Bruker® Daltonics).

2.4 Phenotypic Screening for ESBL-Ec and Carbapenemase Production

All isolated *E. coli* were screened for presumptive ESBL production using CHROMagar™ ESBL medium and MacConkey agar supplemented with 4.0 µg/mL ceftriaxone (Sigma—Aldrich®, Massachusetts, EUA) (MC + CRO), according to Jacob et al. (2020). Additionally, *E. coli* isolates were investigated for the presumptive production of *Klebsiella pneumoniae* carbapenemase (KPC) using CHROMagar™ KPC medium and for carbapenemases using mSuperCARBA medium (Plastlabor) (Gilad et al., 2011; Segarra et al., 2018).

Briefly, 3-µL aliquots of the selected strains, adjusted to a 0.5 McFarland nephelometric standard (approximately 1.5×10^8 CFU/mL) were spot-inoculated on the surface of the selective media. To validate the tests, *E. coli* ATCC 25922 was used as a negative control for ESBL and KPC production, and *E. coli* DH5-α was used as a negative control for carbapenemase; both were obtained from the LEMA/UFF culture collection. *Enterobacter* sp. 849, a carbapenemase producer, was used as a positive control for metallo-β-lactamases, and the ESBL-Ec j609 strain was used as a positive control for ESBL. The two β-lactamase-producing control strains were obtained from the Medical Microbiology Research Laboratory at the Federal University of Rio de Janeiro (LIMM/UFRJ, RJ, Brazil) culture collection.

2.5 Phenotypic Detection of ESBL Production

We investigated all strains that exhibited growth in CHROMagar™ ESBL and MC + CRO, screening simultaneously for specific ESBL production using the double-disk synergy test (DDST) (Jarlier et al., 1988). The same ESBL-positive (*E. coli* j609) and negative (*E. coli* ATCC 25922) strains were used to validate this phenotypic assay.

2.6 Antimicrobial Susceptibility Testing

Escherichia coli isolates were subjected to antimicrobial susceptibility tests using the disk diffusion method on Mueller—Hinton (MH) agar plates (Kasvi) in accordance with CLSI guidelines (Clinical & Laboratory Standards Institute, 2025). The following twenty-two antimicrobial agents (Cecon, São Paulo, SP, Brazil) belonging to seven different classes (aminoglycosides, β-lactams, folate pathway antagonists, nitrofurans, phenicols, quinolones/fluoroquinolones, and tetracyclines) were tested: amikacin 30 µg (AMK), ampicillin 10 µg (AMP), aztreonam 30 µg (ATM), cefazolin 30 µg (CFZ), cefepime 30 µg (FEP), cefotaxime 30 µg (CTX), cefoxitin 30 µg (FOX), ceftazidime 30 µg (CAZ), ceftriaxone 30 µg (CRO), cefuroxime 30 µg (CXM), chloramphenicol 30 µg (CHL), ciprofloxacin 5 µg (CIP), ertapenem 10 µg (ETP), gentamicin 10 µg (GEN), imipenem 10 µg (IPM), levofloxacin 5 µg (LVX), meropenem 10 µg (MEM), nalidixic acid 30 µg (NAL), nitrofurantoin 300 µg (NIT), piperacillin-tazobactam 100/10 µg (TZP), trimethoprim-sulfamethoxazole

1.25/23.75 µg (SXT), and tetracycline 30 µg (TET). The *E. coli* ATCC 25922 strain was used as a quality control for the test.

Isolates were classified as susceptible, susceptible dose dependent, intermediate, or resistant according to CLSI 2025 guidelines. For multidrug resistance (MDR) analysis, isolates that were resistant or intermediate to a given antimicrobial agent were classified as “nonsusceptible”, MDR profile was defined as nonsusceptibility to at least one agent in three or more antimicrobial agent classes (Magiorakos et al., 2012). The multiple antimicrobial resistance (MAR) index per isolate was calculated using the MAR formula, which is applied to a single isolate: a/b . In this formula, a represents the number of antimicrobial agents to which the isolate was resistant, and b represents the total number of antimicrobial agents to which the isolate was exposed (Krumperman, 1983).

2.7 Detection of Antimicrobial Resistance and Integron Integrase Genes

To evaluate possible anthropogenic origins, we used PCR to investigate the presence of clinically relevant ESBL genes ($bla_{CTX-M-1,2}$, $bla_{CTX-M-8}$, $bla_{CTX-M-14}$, bla_{TEM} , and bla_{SHV}) (Campana et al., 2017; da Costa et al., 2020), carbapenem resistance gene (bla_{KPC}) (Hossain et al., 2004), quinolone resistance genes ($qnrA$, $qnrD$ and $qnrS$) (Kraychete et al., 2016), and integron integrase genes ($intI1$, $intI2$ and $intI3$) (Richter et al., 2020; Xu et al., 2008). First, genomic DNA was obtained from each potentially ESBL-Ec isolate by thermal lysis in 300 µL of 5% Chelex 100® (Bio-Rad Laboratories, California, USA), as described by Walsh et al. (2013), with modifications. PCR amplifications were performed using a T100™ thermal cycler (Bio-Rad, Hercules, USA). Each reaction consisted of 50% (v/v) GoTaq® G2 Green Master Mix 2× (Promega, Madison, Wisconsin, USA), 0.8% (v/v) BSA 50 mg/mL (Sigma, USA), 4.8% (v/v) Igepal 1% (Sigma), 200–400 nM of each primer, 10 ng of DNA template, and nuclease-free distilled water. Primers sequences and amplification conditions are detailed in Supplementary Table 2. The strains used as positive controls are shown in Supplementary Table 3. The PCR products were subsequently subjected to 1.0% (w/v) agarose gel electrophoresis.

2.8 Statistical Analyses

Statistical analyses were conducted using R version 4.3.3 (R Core Team, 2024) through the RStudio interface (version 2024.04.0+735, Posit Team, 2024). To analyse the microbiological quality of Jurujuba’s water on the basis of CONAMA Guideline No. 357/2005 (for waters used in bivalve mollusks cultivation for human consumption), the geometric mean and the 90th percentile of 15 thermotolerant coliform MPN/100 mL samples were calculated. Spearman’s correlation coefficient was used to evaluate the association between the MPN of thermotolerant coliforms and physical–chemical parameters. Correlations with a p value ≤ 0.05 were considered statistically significant. Heatmaps were generated using the “ggplot2” (Wickham, 2016), “reshape2” (Wickham, 2007), and “dplyr” (Version 1.1.4) packages to evaluate the antimicrobial nonsusceptibility phenotypic profile and the frequency of ARGs and integrons across sampling sites and time points (Wickham et al., 2023). Network analysis of co-occurrence between $intI1$ and ARGs, as well as between $qnrS$ and other ARGs or $intI1$ were performed using the “igraph” package (version 2.1.4; Csárdi et al., 2025).

3 Results and Discussion

The concept that antimicrobial resistance is a global ecological issue, not limited solely to human clinical settings, is now well established. Although antimicrobial resistance is not a phenomenon resulting exclusively from human activities, the overuse and improper disposal of these drugs, combined with other anthropogenic activities such as livestock farming, agriculture, and the discharge of wastewater into natural ecosystems, have exerted selective pressure that led to the emergence of resistant microorganisms (Zhang et al., 2025b). The rising rate of multidrug-resistant pathogens is alarming, with detection across different sectors, sources and countries. Molecular epidemiological studies describing the circulation of resistant bacterial pathogens are crucial to push for public policies that enable the containment of the spread of this situation, especially related to animal/livestock surveillance (Acharya et al., 2025; Ali et al., 2025; Saeed et al., 2025). Majumdar and co-workers (2026) observed that there is a gap in primary surveillance data from animal/livestock and food-chain, and integrated One Health surveillance.

However, despite the large amount of primary surveillance data from environmental and wastewater sources, there was low detection of policy documents or guidelines and a very limited number of systematic reviews on surveillance in this sector (Majumdar et al., 2026). Consequently, given the crucial role the environment plays as a reservoir and dissemination platform ARGs, conducting molecular epidemiological studies in this sector is important to protect human, animal, and ecosystem health.

Estuaries and shore zones are under continued anthropogenic pressure from economic activities and population growth and deserve more attention for surveillance purposes from a One Health perspective. Especially in densely inhabited regions, these areas are important resistome hotspots that play a decisive role in the propagation of ARGs across indigenous microbial communities and potential pathogens via horizontal gene transfer (Mourão et al., 2025; Zhang et al., 2025a, b). ARG circulation in beach waters has been increasingly documented in Latin America over the last two decades, with Brazil having the highest number of reports and *Escherichia* tied with *Vibrio* as the most frequently studied genera (Montiel-Mora et al., 2026; Costa et al., 2026). To the best of our knowledge, our study is the first to analyse recreational waters in Niterói city (RJ, Brazil) under the Tricycle Surveillance Protocol.

3.1 Evaluation of Physical–chemical Parameters and Bathing Water Quality

The findings from the physical–chemical parameters observed at Jurujuba, Icaraí, and Piratininga during the study period are summarized in Table 1. The details can be found in Supplementary Fig. 1a. Despite substantial variability in individual measurements, mean values were comparable across the sampling sites.

No statistically significant correlations ($p > 0.05$) were observed between physical–chemical parameters (pH, temperature, and salinity) and bathing water quality, indicating that abiotic parameters and microbiological features were not even associated (Table 2). The lack of temperature data for Jurujuba precluded the correlation analysis for this specific parameter. Furthermore, the application of the median value for imputing missing data resulted in a variable devoid of variations, thereby precluding an estimation of the correlation between the MPN and temperature values.

The microbiological analysis of Jurujuba mariculture waters, which was based on 15 random samples obtained during the study period, revealed a geometric mean value of 11.9 MPN/100 mL and a 90th percentile of 248 MPN/100 mL for thermotolerant coliforms (Supplementary Table 4). While the geometric

Table 1 Maximum, minimum, and average values of chemical and physical parameters evaluated at the sampling sites

Parameter	Icaraí			Piratininga			Jurujuba		
	Min. ^a	Max. ^b	Average $\pm$ SD [*]	Min. ^a	Max. ^b	Average $\pm$ SD [*]	Min. ^a	Max. ^b	Average $\pm$ SD [*]
Salinity (‰)	21	35	32.5 $\pm$ 4	32	36	34.9 $\pm$ 1.1	30	40	33.6 $\pm$ 2.9
Water temperature (°C)	20	28	23.9 $\pm$ 1.9	21	26	22.6 $\pm$ 1.2	20	24	20.2 $\pm$ 1
pH	6	7.5	6.9 $\pm$ 0.4	6.5	7.5	6.9 $\pm$ 0.4	6	7.5	6.7 $\pm$ 0.4

^aMinimum value; ^bMaximum value, ^{*} Average $\pm$ Standard Deviation

Table 2 Spearman's correlation coefficients (ρ) and p values between environmental parameters and thermotolerant coliforms (MPN/100 mL) at Icaraí, Jurujuba and Piratininga n/a = not applicable because of the constant variable, MPN—most probable number

Environmental parameters	Spearman's coefficient/ p value	MPN/100 mL		
		Icaraí	Piratininga	Jurujuba
pH	Spearman's coefficient	−0.012	−0.532	0.063
	p value (two-tailed)	0.965	0.061	0.818
Temperature	Spearman's coefficient	0.193	−0.185	n/a
	p value (two-tailed)	0.491	0.544	n/a
Salinity	Spearman's coefficient	−0.053	0.232	−0.117
	p value (two-tailed)	0.852	0.445	0.666

mean value satisfies the criteria established by CONAMA resolution No. 357/2005 for bivalve mollusks cultivation intended for human consumption, the 90th percentile exceeds the regulatory threshold. Otherwise, thermotolerant coliform analyses from mussels remained satisfactory for human consumption, according to regulatory guidelines (Instrução Normativa No. 161/2022; Ministério da Saúde, 2022), in line with the parameters for fish, a specific category of foods not consumed raw, in which the limit is defined as 5×10^2 *E. coli*/g.

Previous studies in the region have reported less favorable conditions. Jayme et al. (2016) examined *P. perna* bivalve mollusks obtained from Itaipu Beach seawater, another ocean beach in Niterói, near the collection site, as well as mollusks commercialized in a fish market in the same city during the years 2012 and 2013. The authors reported that the geometric mean of the water also exceeded 43 MPN/100 mL, which renders it unsuitable for cultivating mussels intended for human consumption. The majority of the mollusks were determined to be unsuitable for human consumption, likely due to post-harvest manipulation. In contrast, we evaluated fresh mussels directly from mariculture, and our analyses revealed that they were satisfactory for consumption.

3.2 Isolation and Identification of Bacteria from Water and Mussels

All the isolates presumably identified as *E. coli* were subjected to MALDI-TOF MS, and only for those with a log score of ≥ 2.00 were considered reliably identified at species level. Thus, 960 out of the 1,061 stored isolates were confirmed as lactose fermenters; of these, 316 isolates were identified as *E. coli*, but only 299 isolates with a score ≥ 2.00 . Most isolates were obtained between June and October/2022; in 2023, we observed a more heterogeneous distribution (Fig. 2a).

The highest percentage of *E. coli* isolates (13.4%, $n=40$) were recovered in February 2023, during summer season, while the lowest percentage of isolated samples (0.3%, $n=1$) was observed in June 2023, when winter began (Fig. 2a). In December 2022, no *E. coli* was isolated, likely due to sampling restricted to Jurujuba. An analysis of the distribution of *E. coli* among the sampling points (Supplementary

Fig. 1b) revealed no clear influence of water quality related to sewage effluent, but it was possible to observe a slight influence of anthropogenic factors in general, especially in the samples from Icaraí beach (42.1%, $n=126$), the most populous municipality evaluated (Fig. 2b). In contrast, from Piratininga, 63 strains were identified as *E. coli* (21.1%), which was the lowest number of isolates from water samples; 98 strains (32.8%) were isolated from Jurujuba (Fig. 2b). The number of *E. coli* isolates from the mussels was the lowest (4%, $n=12$).

Despite the better water quality throughout the period compared with Jurujuba, Icaraí exhibited a greater number of *E. coli* isolates (Fig. 2b), suggesting that there was no direct correlation between the bathing water quality parameters and the survival of *E. coli* in this environment. In site-by-site evaluation, Icaraí had the greatest number of isolates in September 2022, when it was classified as unsuitable. However, when all the sampling sites were unsuitable for bathing, that is, when the MPN value was high, there was a low number of *E. coli* isolates, corroborating the absence of correlation between thermotolerant coliforms and *E. coli* isolation or any of the physical–chemical parameters, since differences observed in each parameter also had no correlation with the number of bacteria isolated.

It is widely demonstrated that bacterial diversity, as well as indigenous resistomes, is influenced by seasonal variations in the water column, fluctuations in salinity, and geographical locations, both in coastal and pristine open ocean waters, from tropical to polar regions (Bunse & Pinhassi, 2017; Dželalija et al., 2023; Serrana et al., 2025). In addition, no correlation could be established in our study; the nature of the different environments studied could explain some of the findings. Although Piratininga has moderate urban activity, it is an open-ocean beach where wave action may inhibit the persistence of such bacteria. A comparison of the findings at Jurujuba and Icaraí revealed that although both beaches are situated within Guanabara Bay, in Icaraí, the population and their pets frequently visit the beach for sports and leisure activities, both on the boardwalk and on the sand, whereas in Jurujuba, the human influence is more related to untreated sewage. In addition, at the Jurujuba sampling site, water was collected in the mariculture zone, not on the beach.

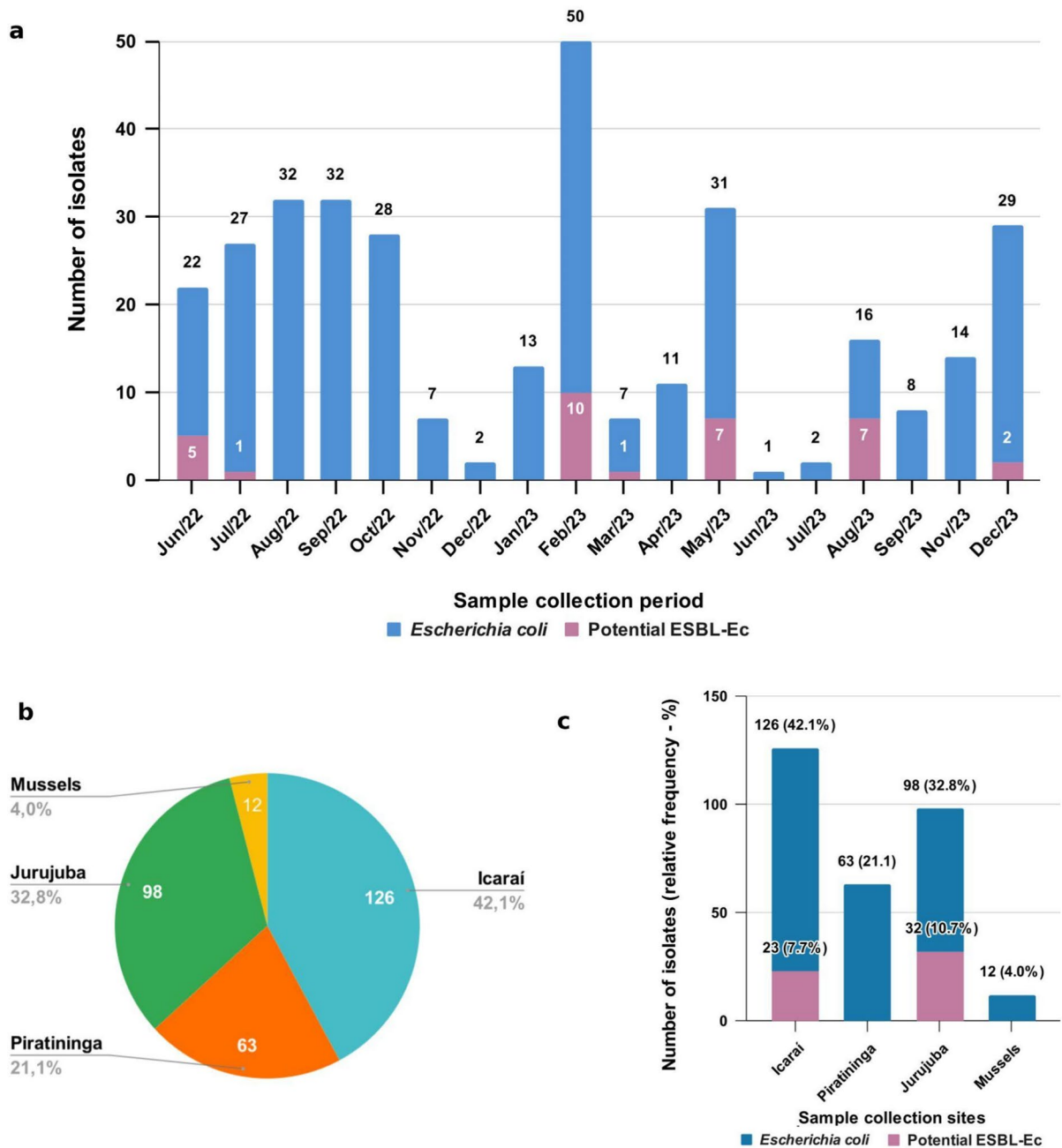


Fig. 2 Distributions of all *E. coli* and ESBL-Ec strains isolated from seawater samples from Jurujuba, Icaraí, and Piratininga (Niterói, RJ) (a) throughout the study period, (b) among the different origins and (c) by sampling site

3.3 Phenotypic Screening for ESBL-Ec and Carbapenemase Production

Phenotypic screening for ESBL-Ec and carbapenemase production was performed on all 299 isolates

identified as *E. coli* (Supplementary Fig. 2). Among the isolates evaluated, 33 (11%) grew on MC + CRO (4.0 µg/mL), 57 (19.1%) grew on CHROMagar™ ESBL medium, and none grew on either CHROMagar™ KPC or CHROMagar™ mSuperCARBA

medium. All 33 isolates that grew on MC + CRO also grew on CHROMagar™ ESBL, and most of them were recovered from Jurujuba's mariculture water (54.5%, $n=18$), followed by Icaraí (42.4%, $n=14$). A single isolate (3.0%) was recovered from Piratininga beach waters (Fig. 2c).

We did not observe any correlation between the number of *E. coli* isolates and ESBL-Ec. Although the greatest number of *E. coli* isolates was recovered in February 23, which included the major ESBL-Ec isolates, August and September 2022 had no ESBL-Ec isolates (Fig. 2a). Although we did not observe a direct correlation with fecal contamination, most ESBL-Ec isolates were obtained from Icaraí and Jurujuba, both located within Guanabara Bay, subjected to anthropogenic influences (Fig. 2c).

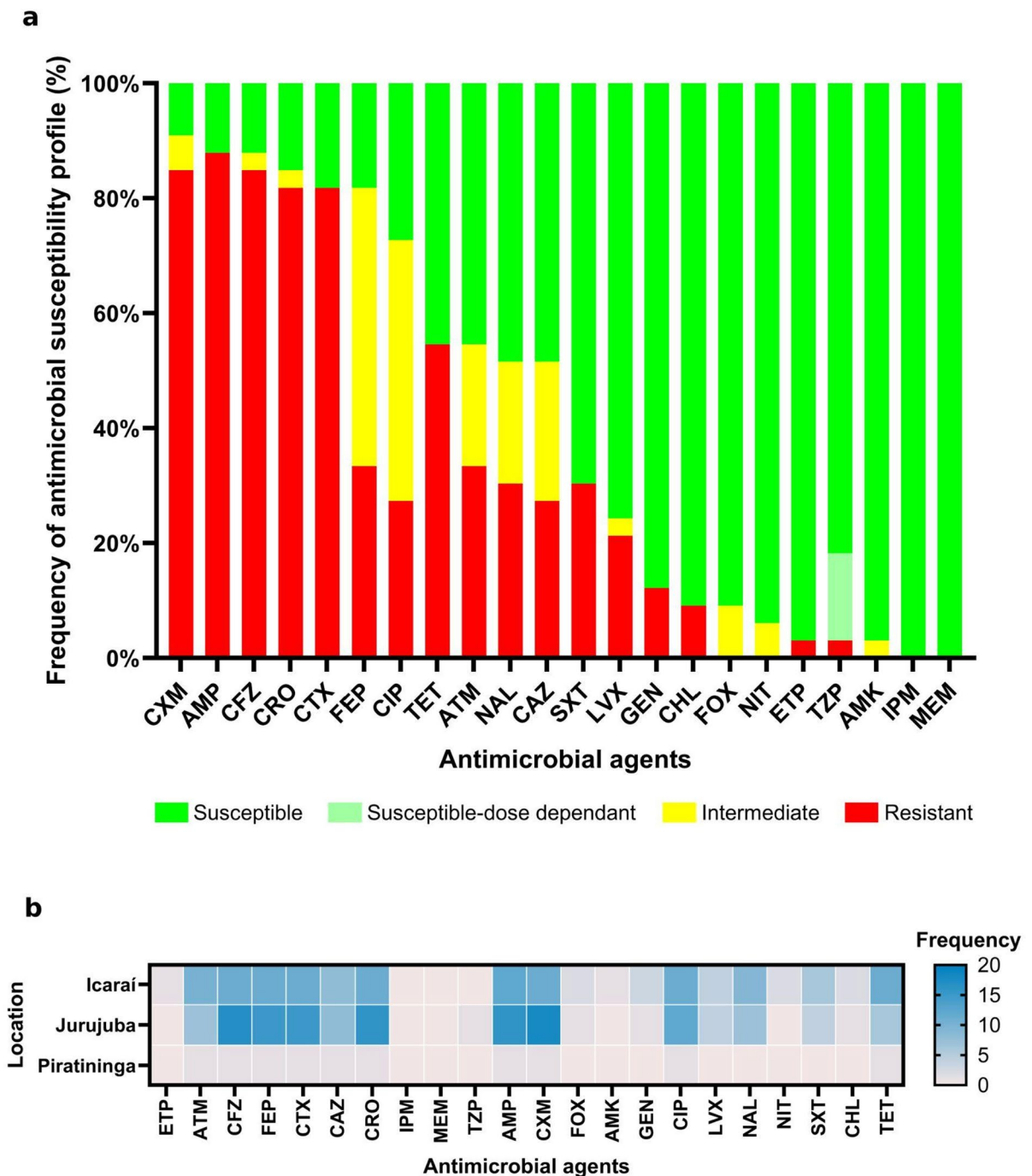
Despite the longstanding utilization of *E. coli* as one of the key indicators of fecal input in water bodies, recent findings suggest that *E. coli* may survive and persist in aquatic environments regardless of the degree of fecal contamination. These "naturalized" strains appear to have coevolved in wastewater and sewage niches, resulting in several adaptations that ensure their survival under a variety of stresses, including enhanced resistance to heat, heavy metals, oxidants, biocides, and even antimicrobial agents (Cho et al., 2020; Yu et al., 2022). In addition, microplastics are hotspots for MDR bacteria and ARGs, facilitating HGT events (Gross et al., 2025; Magalhães et al., 2024; Siddique et al., 2025), and Jurujuba retains macro, meso-, and microplastics in the surface water, bottom, and beach sediments (Castro et al., 2020). Mussel colonies are cultivated around plastic jars and PET bottles (Fig. 1), which can serve not only as sources of microplastics in the water but also as platforms for biofilms. This is likely another key factor influencing the persistence of these bacteria in the region. However, importantly, no ESBL-producing *E. coli* were isolated from the mussels, and no evidence of fecal contamination was detected in these animals, despite the observed water quality conditions.

3.4 Antimicrobial Susceptibility Profile

All 33 presumptive ESBL-Ec isolates were subjected to antimicrobial susceptibility testing. High percentages of nonsusceptibility (intermediate plus

resistant phenotypes) were detected for cefuroxime (90.9%, $n=30$), cefazolin and ampicillin (87.9%, $n=29$), and ceftriaxone (84.8%, $n=28$) (Fig. 3a, b). The high frequency of nonsusceptibility to ampicillin was expected, given its classification as a penicillin subclass, the first β -lactam subclass described (Bush, 2023). The predominance of the nonsusceptibility profile is consistent with the findings of Andrade et al. (2015), who reported a high frequency of ampicillin resistance in *E. coli* isolates obtained from beaches in Ilha Porchat and Gonzaguinha, São Vicente (São Paulo, Brazil). Although located far from the sampling sites of the present study, these data show the persistence of these phenotypes in marine ecosystems.

In contrast, the lowest nonsusceptibility frequencies were observed for ertapenem, piperacillin-tazobactam, and amikacin (3.0%, $n=1$), followed by nitrofurantoin (6.1%, $n=2$) and chloramphenicol (9.1%, $n=3$) (Fig. 3a). All the isolates were susceptible to imipenem and meropenem. Among β -lactams used to treat MDR gram-negative bacterial infections in hospital settings, carbapenems represent the last-resort antimicrobial agents (Agarwal et al., 2023; Bush, 2013; Lermينياux et al., 2023). In veterinary medicine, the use of carbapenem drugs is not recommended; however, some countries still use these drugs to treat complicated infections in animals (Sellera et al., 2021). The elevated carbapenem susceptibility observed in this study may indicate that the environment has not yet been affected by the dissemination of carbapenem resistance genes. However, in a study by Montezzi et al. (2015), the *bla*_{KPC-2} gene was found in *Citrobacter* spp., *Kluyvera* spp., *Enterobacter* spp., and *Aeromonas* spp. isolated from seawater samples in the city of Rio de Janeiro, specifically from the beaches of Flamengo and Botafogo, which are located within Guanabara Bay and are frequently classified as unsuitable for bathing by INEA. Nonetheless, recreational activities persist in these locations. Several studies conducted in the metropolitan region of Rio de Janeiro have identified carbapenem resistance genes in microorganisms isolated from aquatic environments (de Araujo et al., 2016; Paschoal et al., 2017), in *E. coli* and *K. pneumoniae* isolated from surveillance cultures of hospitalized patients (Lima et al., 2021), and in *E. coli* isolated from houseflies captured near hospitals in the city of Rio de Janeiro (Carramaschi et al., 2023).



The detection of carbapenem resistance genes in bacteria of aquatic origin underscores the role of these environments as reservoirs of these genes and as vehicles for carbapenem-resistant microorganisms. In this context, monitoring these genes in marine environments is imperative, given the potential for their

dissemination to the community, which is likely to occur at an unprecedented rate in coastal and estuarine regions.

As illustrated in Fig. 3b, the heatmap shows that isolates originating from Jurujuba waters had variable frequencies of nonsusceptibility to several

◀ **Fig. 3** (a) Distribution of antimicrobial susceptibility profiles among *E. coli* strains isolated in the study. The bars represent the relative frequency (%) of the susceptibility categories for each antimicrobial agent, according to CLSI guidelines (2025): susceptible (green box), susceptible-dose dependent (light green box), intermediate (yellow box), and resistant (red box). Antimicrobial agents are ordered from highest to lowest resistance frequency. CXM, cefuroxime; AMP, ampicillin; CFZ, cefazolin; CRO, ceftriaxone; CTX, cefotaxime; FEP, cefepime; CIP, ciprofloxacin; TET, tetracycline; ATM, aztreonam; NAL, nalidixic acid, CAZ, ceftazidime; SXT, sulfamethoxazole-trimethoprim; LVX, levofloxacin; GEN, gentamicin; CHL, chloramphenicol; FOX, cefoxitin; NIT, nitrofurantoin; ETP, ertapenem; TZP, piperacillin-tazobactam; AMK, amikacin; IPM, imipenem; MEM, meropenem. (b) Heatmap of the antimicrobial nonsusceptibility phenotype of *E. coli* strains, showing the distribution by location

β -lactam antimicrobial agents, including ampicillin, cefazolin, cefepime, ceftriaxone, cefuroxime, and cefotaxime. The relative distribution of the nonsusceptibility phenotypes was heterogeneous across the sampling locations (Supplementary Fig. 3). Nevertheless, Jurujuba had a higher frequency of nonsusceptible strains for β -lactam and other antimicrobial agent classes. In Piratininga, a single nonsusceptible isolate was detected (3.0%), exhibiting resistance to ten distinct antimicrobial agents (Supplementary Fig. 3). ESBL genes confer resistance to monobactams, penicillins, and third-generation cephalosporins while remaining susceptible to carbapenems and β -lactamase inhibitors (Macesic et al., 2025). In this study, a high frequency of resistance to third-generation cephalosporins, including cefotaxime, ceftriaxone, and ceftazidime, was observed, supporting the presence and expression of ESBL genes among the majority of the isolates.

Among non- β -lactam antimicrobial agents, nonsusceptibility of ESBL-Ec isolates for ciprofloxacin, tetracycline, and nalidixic acid was prominent (Fig. 3a, b; Supplementary Fig. 3). Tetracycline and quinolone/fluoroquinolone resistance in *E. coli*, especially ciprofloxacin resistance, is a global issue that has been detected over the past 20 years despite efforts to contain it (Karami et al., 2006; Tchesnokova et al., 2023). In particular, the resistance of ExPEC/UPEC strains to ciprofloxacin is increasing, even though the use of this antibiotic has been discouraged and is associated with cephalosporin resistance (Tchesnokova et al., 2023). Therefore, the findings concerning the co-occurrence of ciprofloxacin and third-generation cephalosporins in environmental strains

in this work are alarming and raise awareness of this public health issue.

The relatedness between human and poultry *E. coli* isolates was recently evaluated in Egypt, and the study detected 58.6% ESBL producers, of which 96.5% were MDR, mostly isolated from poultry (Ahmed et al., 2025). Chicken isolates showed a higher frequency of fluoroquinolones and quinolones than human isolates, suggesting that selective pressure is greater in poultry farming than in human clinical practice. Virulence genes were also detected, but no significant differences were observed, suggesting a zoonotic route of transmission. However, when evaluating different meat sources (chicken meat and beef by-products) and samples from humans with diarrhoea, the highest rates of resistance were observed in isolates of human origin (El-Hamid et al., 2026). Statistical analysis indicated significant differences in antimicrobial resistance profiles—specifically regarding meropenem, cefotaxime, ceftazidime, ciprofloxacin, and levofloxacin—among *E. coli* isolates from the different sources. In Bangladesh, a comparative study involving ESBL-Ec isolates from urban live-bird markets and rural poultry farms detected a higher abundance of ESBL-Ec in urban live-bird markets, as well as in dead birds compared to live ones (Amin et al., 2025). Although this scenario worldwide, a study on oysters in Egypt detected a low frequency of quinolone resistance, despite the detection of MDR *E. coli* in over 65% of the isolates (Mohammed et al., 2024). When analysing virulence potential, it was observed that most isolates were related to ExPEC and EPEC pathotypes (87.5% and 70.8%, respectively). Here, all 33 MDR isolates were resistant to ampicillin and tetracycline (Fig. 4). These isolates exhibited 15 distinct resistotypes (Fig. 4, Table 3). Among the MDR *E. coli* isolates, in addition to a high frequency of nonsusceptibility to quinolones, frequent nonsusceptibility to tetracyclines (tetracycline) and sulfonamides plus diaminopyrimidine (sulfamethoxazole/trimethoprim) was also observed. These findings are particularly important, given that these three classes of antimicrobial agents are used to treat bacterial infections in both human and animal subjects (LaPlante et al., 2022; Poirer et al., 2018).

It has been observed and reported that there are differences in abundance and variety of antimicrobial resistance profiles, even regarding ESBL-Ec,

Fig. 4 Overview of the phenotypic and genotypic test results for the 33 *E. coli* strains isolated in the study

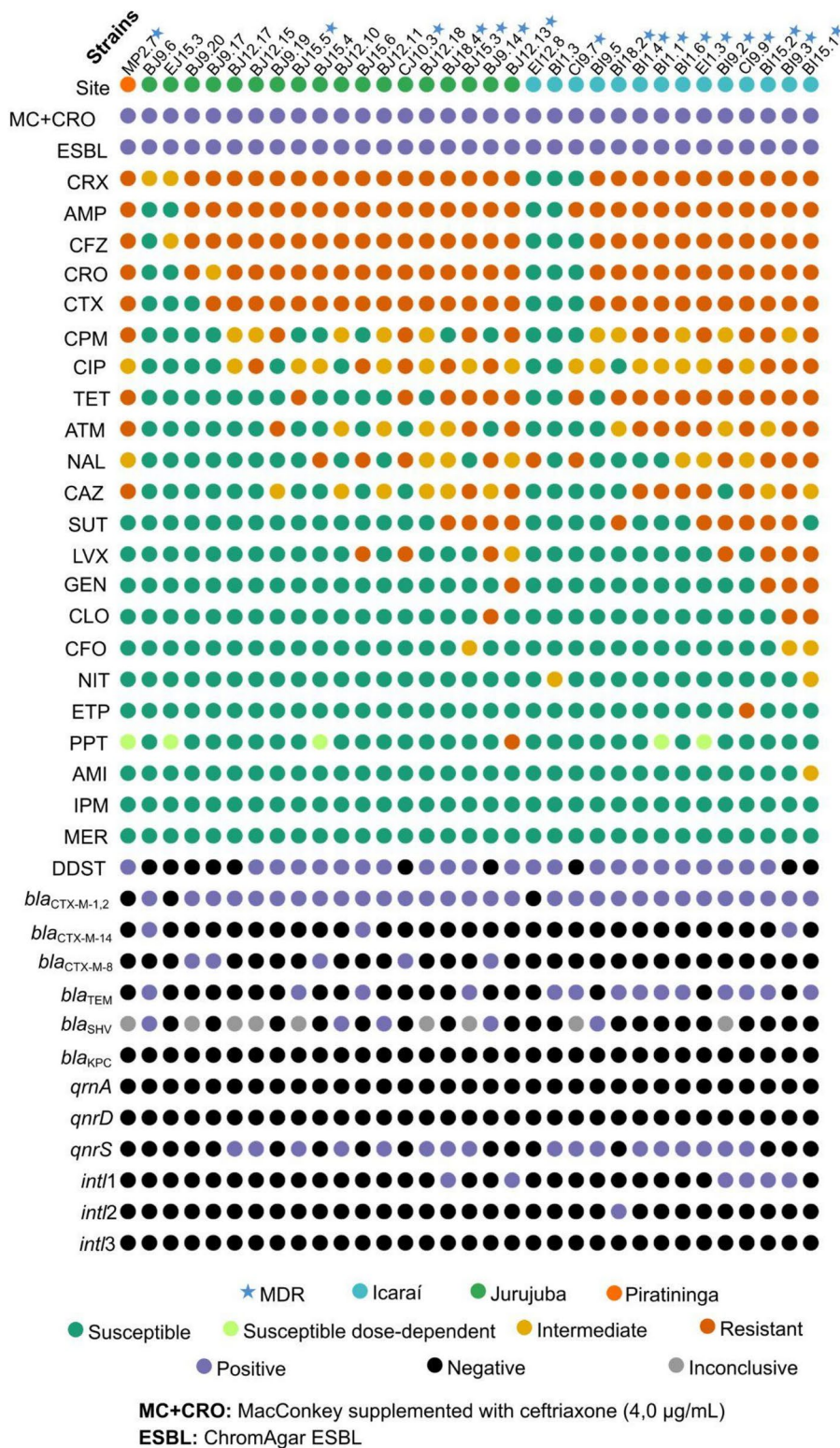


Table 3 Overview of the 33 MDR *E. coli* strains investigated in the study

Strain	Sampling	Site ^a	Screening ^b	Antimicrobial agents ^c	MAR index ^a	ESBL phenotype	Genes
BI15.1	15	I	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, FOX, AMK, GEN, CIP, LVX, NAL, NIT, CHL, TET	0.8	—	<i>bla</i> _{CTX-M-1,2}
BI9.3	9	I	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, FOX, GEN, CIP, LVX, NAL, SXT, CHL, TET	0.7	—	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{CTX-M-14+} , <i>intl1</i>
BI15.2	15	I	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, GEN, CIP, LVX, NAL, SXT, TET	0.6	+	<i>bla</i> _{CTX-M-1,2} , <i>intl1</i>
CI9.9	9	I	MC + CRO, ESBL	ETP, ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, CIP, NAL, SXT, TET	0.6	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{TEM+} , <i>qnrS</i> , <i>intl1</i>
BI9.2	9	I	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CRO, AMP, CXM, CIP, LVX, NAL, SXT, TET	0.5	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{TEM+} , <i>qnrS</i> , <i>intl1</i>
EI1.3^d	1	I	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, CIP, NAL, SXT, TET	0.5	+	<i>bla</i> _{CTX-M-1,2} , <i>qnrS</i>
BI1.6^e	1	I	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, CIP, NAL, SXT, TET	0.5	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{TEM+} , <i>qnrS</i>
BI1.1^f	1	I	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, CIP, TET	0.5	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{TEM+} , <i>qnrS</i>
BI1.4^f	1	I	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, CIP, TET	0.5	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{TEM+} , <i>qnrS</i>
BI18.2	18	I	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CRO, AMP, CXM, SXT, TET	0.5	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{TEM+} , <i>intl2</i>
BI9.5	9	I	MC + CRO, ESBL	CFZ, FEP, CTX, CRO, AMP, CXM, CIP	0.3	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{SHV+} , <i>qnrS</i>
CI9.7	9	I	MC + CRO, ESBL	AMP, CIP, NAL, TET	0.2	—	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{TEM+} , <i>qnrS</i>
BI1.3	1	I	MC + CRO, ESBL	NIT	0.0	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{TEM+} , <i>qnrS</i>
EI12.8	12	I	MC + CRO, ESBL	NAL	0.0	+	—
BJ12.13	12	J	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, TZP, AMP, CXM, GEN, CIP, LVX, NAL, SXT, TET	0.7	+	<i>bla</i> _{CTX-M-1,2} , <i>intl1</i>

Table 3 (continued)

Strain	Sampling	Site ^a	Screening ^b	Antimicrobial agents ^c	MAR index ^a	ESBL phenotype	Genes
BJ9.14	9	J	MC + CRO, ESBL	CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, CIP, LVX, NAL, SXT, CHL, TET	0.6	—	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{CTX-M-8} , <i>bla</i> _{SHV}
BJ15.3	15	J	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, FOX, CIP, SXT, TET	0.5	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{TEM} , <i>qnrS</i>
BJ18.4^d	18	J	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, CIP, NAL, SXT, TET	0.5	+	<i>bla</i> _{CTX-M-1,2} , <i>qnrS</i> , <i>int11</i>
BJ12.18	12	J	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, CIP, NAL	0.5	+	<i>bla</i> _{CTX-M-1,2} , <i>qnrS</i>
CJ10.3	10	J	MC + CRO, ESBL	CFZ, FEP, CTX, CRO, AMP, CXM, CIP, LVX, NAL, TET	0.5	—	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{CTX-M-8}
BJ12.11	12	J	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, CIP	0.4	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{SHV} , <i>qnrS</i>
BJ15.6	15	J	MC + CRO, ESBL	CFZ, FEP, CTX, CRO, AMP, CXM, CIP, LVX, NAL	0.4	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{CTX-M-14} , <i>bla</i> _{TEM}
BJ12.10	12	J	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM	0.4	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{SHV} , <i>qnrS</i>
BJ15.4	15	J	MC + CRO, ESBL	CFZ, FEP, CTX, CRO, AMP, CXM, CIP, NAL	0.4	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{CTX-M-8}
BJ15.5	15	J	MC + CRO, ESBL	CFZ, FEP, CTX, CRO, AMP, CXM, CIP, TET	0.4	+	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{TEM} , <i>qnrS</i>
BJ9.19	9	J	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM	0.4	+	<i>bla</i> _{CTX-M-1,2}
BJ12.15	12	J	MC + CRO, ESBL	CFZ, FEP, CTX, CRO, AMP, CXM, CIP	0.3	+	<i>bla</i> _{CTX-M-1,2} , <i>qnrS</i>
BJ12.17	12	J	MC + CRO, ESBL	CFZ, FEP, CTX, CRO, AMP, CXM, CIP	0.3	—	<i>bla</i> _{CTX-M-1,2} , <i>qnrS</i>
BJ9.17	9	J	MC + CRO, ESBL	CFZ, FEP, CTX, CRO, AMP, CXM	0.3	—	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{CTX-M-8}
BJ9.20	9	J	MC + CRO, ESBL	CFZ, CRO, AMP, CXM	0.2	—	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{CTX-M-8}
EJ15.3	15	J	MC + CRO, ESBL	CFZ, CXM	0.1	—	—
BJ9.6	9	J	MC + CRO, ESBL	CXM	0.0	—	<i>bla</i> _{CTX-M-1,2} , <i>bla</i> _{CTX-M-14} , <i>bla</i> _{TEM} , <i>bla</i> _{SHV}

Table 3 (continued)

Strain	Sampling	Site ^a	Screening ^b	Antimicrobial agents ^c	MAR index ^d	ESBL phenotype	Genes
MP2.7^e	2	P	MC + CRO, ESBL	ATM, CFZ, FEP, CTX, CAZ, CRO, AMP, CXM, CIP, NAL, TET	0.5	+	—

^aI—Icarai, J—Jurujuba, P—Piratiníngua; ^bMC + CRO—MacConkey agar supplemented with ceftriaxone (4.0 µg/mL), ESBL—CHROMagar™ ESBL; ^cAMP—ampicillin, TZP—piperacillin + tazobactam, CFZ—cefazolin, CXM—cefuroxime, FOX—cefoxitin, CTX—cefotaxime, CAZ—ceftazidime, CRO—ceftriaxone, FEP—cefepime, A™—aztreonam, ETP—ertapenem, IPM—imipenem, MEM—meropenem, AMK—amikacin, GEN—gentamicin, SXT—sulfamethoxazole + trimethoprim, CHL—chloramphenicol, NIT—nitrofurantoin, NAL—nalidixic acid, CIP—ciprofloxacin, LVX—levofloxacin, TET—tetracycline; ^dThe multiple antimicrobial resistance (MAR) index was calculated using the formula = *a/b*, where *a* represents the number of antimicrobial agents to which the isolate was resistant and *b* represents the number of antimicrobial agents to which the isolate was exposed (Krumperman, 1983); ^eResistotype: ATM-CFZ-FEP-CTX-CAZ-CRO-AMP-CXM-CIP-NAL-SXT-TET; ^fResistotype: ATM-CFZ-FEP-CTX-CAZ-CRO-AMP-CXM-CIP-TET; ^gResistotype: ATM-CFZ-FEP-CTX-CAZ-CRO-AMP-CXM-CIP-TET. Bold indicates the strains classified as MDR, that is, with nonsusceptibility to at least one antimicrobial in three or more distinct classes (Magiorakos et al., 2012)

according to the source of bacterial isolation, origin of the samples, and geographic region of study, that are correlated to the medications administered, their recommendations and regulatory agencies (Beaumont et al., 2026; Karabasil et al., 2025). An observational study with data obtained from Antimicrobial Testing Leadership and Surveillance (ATLAS) project permitted to analyse the incidence of MDR and ESBL-producing *E. coli* from 64 countries (Al-Mustapha et al., 2025). ESBL-Ec incidence data were normalized, and the results showed a significantly higher incidence in isolates from North America and Asia than in those from Africa and South America. *E. coli* is a global indicator pathogen selected to monitor the spread of AMR, yet it may not reflect the bacterial ecology of every country—even though the *E. coli* dataset featured excellent representation through 64 countries.

On the other hand, according to Beaumont et al. (2026), low- and middle-income countries (LMICs) showed a higher prevalence of ESBL-Ec carriers than high-income countries (HICs), likely due to differences in socioeconomic levels (Beaumont et al., 2026). Furthermore, the transmission pattern differs: while transmission in LMICs occurs across environmental, human, and animal sectors, human-to-human transmission predominates in HICs.

MAR indices ranged from 0.0 to 0.8, with an average of 0.4, and a total of 18 (54.4%) isolates exhibited an MDR phenotype, with 11 (33.3%) isolates from Icarai, six (18.2%) from Jurujuba, and one (3.0%) from Piratiníngua. Twenty-seven (81.8%) isolates had MAR values > 0.2 (Table 3). According to the literature, isolates with a MAR above 0.2 represent a high risk, in addition to being associated with environments where antimicrobial agents are frequently administered (Alsufyani et al., 2026; Isaiah et al., 2025). The MAR index of a particular MDR isolate was 0.2, which was attributed to its resistance to four antimicrobial agents, three of which belong to distinct classes.

When analysing ESBL-Ec in meat with human health implication, it was observed that *bla*_{CTX-M} and *intI* genes were more frequent in human isolates than in chicken or beef meat (El-Hamid et al., 2026). MAR indices of the majority of 90 isolates were above 0.64. *Escherichia coli* isolated from clinical and non-clinical samples and, analysing by resistotype, four *E. coli* isolates from chicken meat and human stool samples were resistant to 10 antimicrobial

agents (MAR=0.91). In Egypt, Mohammed et al. (2024) detected 75% of *E. coli* isolated from oysters presented MAR indexes surpassing 0.2, with a peak index of 0.6. In the present study, a higher MAR index of 0.8 was observed in an ESBL-Ec isolate from Icaraí, resistant to 17 antimicrobial agents and six different classes. Furthermore, 84.6% of the strains from Icaraí exhibited an MAR index greater than 0.2. MAR index values close to 1 are associated with exposure to large quantities of antibiotics or, at the very least, continuous selective pressure. Twenty-seven ESBL-Ec strains presented a MAR index above 0.2, characterizing our coastal ecosystem as a highly impacted environment.

The presence of MDR *E. coli* in coastal environments used for recreation and fishing represents a potential risk to both human and animal health, underscoring the importance of continuous environmental surveillance within the multisectoral One Health approach.

3.5 Phenotypic and Genotypic Detection of ESBL

The 33 presumptive ESBL-Ec strains that grew simultaneously on MC+CRO and CHROMagar™ ESBL were subjected to DDST to determine their phenotypic ESBL production capacity. Twenty-two (66.7%) isolates displayed a positive phenotype. ESBL production was detected in 11 (50.0%), 10 (45.5%), and one (4.5%) *E. coli* isolate recovered from Icaraí, Jurujuba, and Piratininga, respectively (Fig. 4, Table 3). The use of CHROMagar™ ESBL and MC+CRO in the initial phenotypic screening proved efficient for the preliminary detection of ESBL-Ec isolates. Most of the isolates screened were confirmed to have phenotypic ESBL production by DDST.

The investigative approach involved screening all 33 presumptive ESBL-Ec strains for the presence of β -lactam resistance genes by PCR, even those with negative DDST results. Among the ESBL genes investigated, *bla*_{CTX-M-1,2} genes were the most frequently detected among the isolates (90.9%, $n=30$), in addition to the integrase gene *intI1* (Fig. 4, Table 3). Among all the genes detected in this study, *bla*_{CTX-M-1,2} genes were detected 30 times (30/79; 38%) (Fig. 5a) and were distributed between Icaraí (35.1%, $n=13$) and Jurujuba (40.5%, $n=17$) (Fig. 5b). Analysis of the 33 strains studied revealed *bla*_{CTX-M-1,2} genes in 94.4% ($n=17$) of the strains

isolated from Jurujuba and in 92.9% ($n=13$) of the strains isolated from Icaraí (Supplementary Fig. 4). The prevalence of *bla*_{CTX-M-1,2} in this study suggests the presence of type 1 or 2 CTX-M alleles, making it necessary to confirm the specific group to enable more detailed analyses of the prevalence of these genes in the region.

In a previous study conducted in Rio de Janeiro, which focused on ESBL-Ec isolated from water samples collected from Guanabara Bay, Costa et al. (2023) reported a predominance of *bla*_{CTX-M-1,2}, suggesting the persistence of this gene in marine environments, which may be related to the continuous discharge of treated and untreated sewage into the bay. In Jurujuba, *bla*_{CTX-M-8} was the most frequently detected gene and was exclusively identified at this site, indicating that this environment has greater diversity of ESBL genes than Icaraí beach does (Fig. 5b). Other studies have also detected the *bla*_{CTX-M-8} gene in *E. coli* strains isolated from polluted mangroves and sewage treatment plants in Brazil (Dropa et al., 2016; Sacramento et al., 2018), reinforcing the association with anthropogenically impacted environments.

Although Icaraí accounted for the highest number of *E. coli* isolates, ESBL-Ec was more prevalent in Jurujuba, where higher MPN values were also recorded. This discrepancy may be explained by site-specific hydrodynamic and environmental characteristics. Jurujuba mariculture area is located in a cove within Guanabara Bay, an area subject to urban pollution from the cities of Niterói and Rio de Janeiro, two well-developed urban centers of Rio de Janeiro state (Paulino, 2021). According to Fries et al. (2019), the Jurujuba cove has a low water circulation velocity, which is related to the accumulation of organic matter, nutrients, and microorganisms. Such conditions likely favor the persistence and proliferation of antimicrobial-resistant bacteria, which may explain the prevalence of ESBL-Ec isolates detected in Jurujuba.

The *bla*_{TEM} gene was the second most frequently detected β -lactam resistance gene (Fig. 5a) and it was predominantly found in isolates from Icaraí (Fig. 5b). This gene is commonly reported in ESBL-Ec and has been reported in several studies investigating β -lactam resistance in diverse bacterial species isolated from marine environments in Rio de Janeiro (Alves et al., 2024; Canellas et al., 2021; Costa et al., 2023; Montezzi et al., 2015). However, the predominance of *bla*_{CTX-M} genes among ESBL-Ec isolates observed in

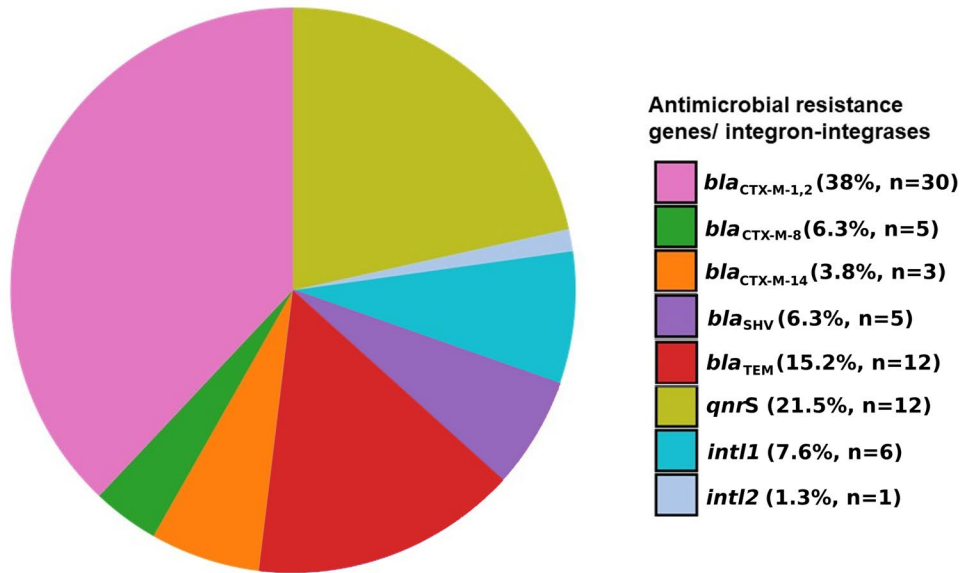
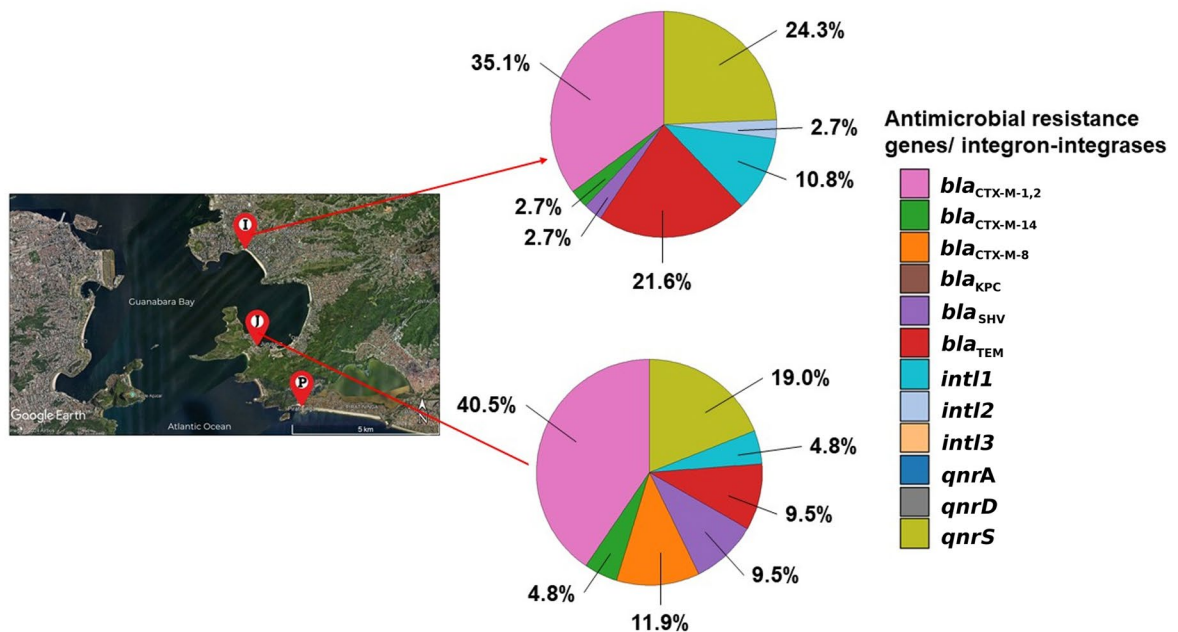
a**b**

Fig. 5 (a) Overall distribution of antimicrobial resistance genes (ARGs) and integrase genes detected in ESBL-Ec strains. (b) Distribution of antimicrobial resistance genes

(ARGs) and integrase genes detected in ESBL-Ec strains isolated from Icarai (I) and Jurujuba (J)

this study is consistent with the findings from multiple sources, including commensal and pathogenic bacteria from human subjects (Ayukafangha et al.,

2026; Carvalho et al., 2021; Ghenea et al., 2022; Muhummed et al., 2024; Park et al., 2024; Rodrigues et al., 2022), animal subjects, both livestock and

wild animals, (Bazalar-Gonzales et al., 2024; de Carvalho et al., 2020; Niharika et al., 2024;), and surface water environments (Beshiru et al., 2024; Costa et al., 2023; Gray et al., 2025; Hassen et al., 2020; Liu et al., 2018), in Brazil and worldwide.

A variety of high-risk clones were reported carrying key resistance genes, especially associated with ESBL dissemination, not only isolated from humans, but also found in animals and environment (Cahyani et al., 2026). Fluoroquinolone resistance and ESBL genes from ST73, a highly pathogenic lineage, frequently isolated from humans and companion animals (Pitout et al., 2023); ST10 carrying CTX-M and other ESBL genes, isolated from humans, livestock and environment (Kocsis et al., 2022); CTX-M-15 dominant ST155 clone, associated to humans, poultry and environment, which supports inter-sector spread (Acharya et al., 2025). Recognizing that different environments, as well animals can play a role as a reservoir of antimicrobial resistance (Zhang et al., 2025b; Doyle et al., 2025; Karabasil et al., 2025; Malcom & Bowes, 2025; Beaumont et al., 2026) is the first step to implement strategies using the One Health approach to control and mitigate the RAM. The development of -omics technologies has enabled the advancement of genomic surveillance, making it possible to monitor the dissemination and persistence of resistance genes. Furthermore, these technologies enable tracking of mutation rates in ARGs within bacterial populations, particularly for monitoring resistant pathogens (Djordjevic et al., 2024). Among the 22 positive strains in DDST, two (10%) did not present any of the β -lactam resistance genes tested. In addition, 10 negative strains in DDST were positive for at least one of the genes tested, indicating that cultivation under selective pressure, both with chromogenic medium for ESBL or MC + CRO, is a powerful screening tool for ESBL-Ec. Most of the 33 presumptive ESBL-Ec strains carried more than one β -lactam resistance gene, with 54.5% ($n=18$) positive for two genes, 6.1% ($n=2$) positive for three genes, and 3.0% ($n=1$) positive for four genes. A single gene was detected in 27.3% ($n=9$) of the strains, while no genes were detected in 9.1% ($n=3$) (Table 3). Despite their importance in clinical practice, phenotypic tests for detecting antimicrobial resistance mechanisms are not

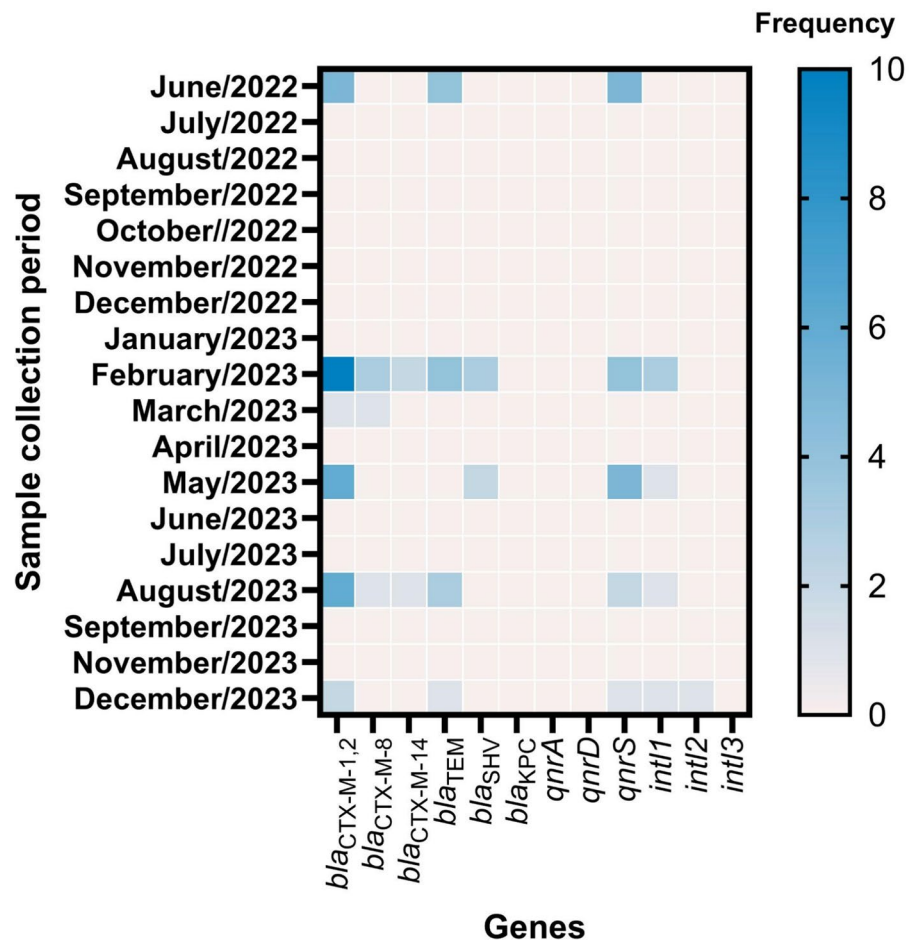
always consistent with the results of molecular investigations. Testing for sensitivity and specificity can lead to differences between genotypic and phenotypic ESBL detection (Castanheira et al., 2021).

In this study, 90.9% of the strains that tested positive in the phenotypic ESBL detection test were confirmed molecularly. One strain (MP 2.7), the unique isolated from Piratininga, was among the positive *E. coli* isolates in the DDST. However, no β -lactamase gene was detected in MP 2.7. Additionally, this isolate was resistant to the third-generation cephalosporins cefotaxime, ceftriaxone, and ceftazidime, which are key indicators for ESBL investigation (Table 3). Giving the diversity of ESBL gene families, the absence of detectable ESBL genes in this isolate may be explained by the presence of other ESBL genes not tested in this study. Indeed, a wide range of ESBL genes, such as *bla*_{CTX-M-15}, *bla*_{CTX-M-9}, *bla*_{GES}, have been reported in other studies that investigated ESBL genes in aquatic environments (Davidova-Gerzova et al., 2023; de Araujo et al., 2016; Montezzi et al., 2015; Paschoal et al., 2017).

3.6 Detection of Additional Antimicrobial Resistance Genes and Integron Integrase Genes

February 2023 had the highest number of *E. coli* investigated in the study, all obtained from Icaraí, and was associated with the greatest genetic diversity during the study. Seven of the eight ARGs identified in the study were detected during this period (summer) (Fig. 6). In addition to the *bla*_{CTX-M-1,2} genes, which were predominant, the *qnrS* gene was detected in all the isolates from June 2022 to May 2023, and the *intI1* gene was detected in the strains isolated on February 2023 (Fig. 6). The months from December to March are summer occurs in Brazil. Summer in the Southeast Region is characterized by elevated temperatures, high humidity and frequent rainfall. Other factors, such as cold fronts, the South Atlantic Convergence Zone, the South Atlantic High-Pressure system, and phenomena such as *El Niño* and *La Niña*, can also influence summer rainfall and temperature (Climatempo, 2022). Of note, 2023 was one of the warmest years on record in Brazil, with persistently elevated temperatures, continued to be recorded in 2024 (Instituto Nacional de Meteorologia, 2024, 2025).

Fig. 6 Detection of anti-microbial resistance genes (ARGs) and integrase-encoding genes in *E. coli* strains isolated and selected throughout the collection period in the present study



The *qnrS* gene was the only quinolone resistance genetic determinant detected and was found in 64.3% ($n=9$) of the isolates from Icaraí and 44.4% ($n=8$) of the strains from Jurujuba (Fig. 5b). All the isolates that had the *qnrS* gene also had at least one β -lactam resistance gene (Table 2). The *int1* gene was detected in 28.6% ($n=4$) of the isolates from Icaraí and 11.1% ($n=2$) of the isolates from Jurujuba. The *int2* gene was the least frequent gene in the study and was identified in a single isolate (3.0%) from Icaraí. However, the *bla*_{KPC}, *qnrA*, *qnrD*, and *int3* genes were not detected in any of the isolates analysed. The antimicrobial resistance profiles of the *E. coli* isolates differed significantly across the beaches. Isolates from different beaches presented variations in nonsusceptibility profiles and in the distribution of ARG and integrase genes, reinforcing the heterogeneity of the bacterial populations across locations.

Integrins are important genetic elements that favor the dissemination of ARGs through HGT (Bhat et al., 2023). Among the integron genes investigated, *int1* was the most common, being detected predominantly in isolates from Icaraí. The *int2* gene was detected in a single *E. coli* isolate recovered from Icaraí. Among the five classes of known integron integrase genes, class 1 integrons are the most clinically significant. They are widely recognized for their ability to act as vectors of several ARGs and represent the most frequent integron class observed in gram-negative pathogens (Ghaly et al., 2021; Wachino, 2025). Although class 1 integrons are not mobile, they are often associated with plasmids, which facilitate their horizontal transfer (Pfeifer et al., 2022). This highlights their critical role in the dissemination of ARGs and underscores the importance of continued investigation of these genetic elements.

Additionally, the presence of class 1 integrons in natural environments has been identified as a bio-marker of anthropogenic pollution because of their ability to capture and express exogenous ARGs, disinfectants, and heavy metals (Gillieatt & Coleman, 2024). The low fitness cost of these elements in *E. coli* strains enables bacteria to adapt to the selective pressure exerted by antimicrobial agents, thereby enabling their survival.

Among the 18 MDR isolates recovered in this study, 17 (94,4%) carried the *qnrS* gene, 13 (72,2%) were ESBL producers, and seven isolates harbored at least one integron integrase/encoding gene (Fig. 4, Table 3). ESBL genes have been identified in MDR *E. coli* strains isolated from aquatic sources in several studies (Blaak et al., 2014; Hossain et al., 2025; Satyaprakash et al., 2024). These MDR ESBL-Ec strains can cause serious infections in humans and animals because of their combined potential for antimicrobial resistance (to β -lactams, mainly to third-generation cephalosporins, and to other classes), which makes the treatment of infections caused by these strains more challenging.

Network analysis of *intI1* and *qnrS* in *E. coli* strains revealed a high degree of co-occurrence between *qnrS* and *bla*_{CTX-M-1,2} and a notable co-occurrence between *intI1* and *bla*_{CTX-M-1,2} (Fig. 7).

The presence of ESBL-producing MDR *E. coli* isolates carrying quinolone resistance genes in natural marine environments is of concern, as it draws attention to the dissemination of ARGs in these ecosystems, serving as a possible source of contamination for humans and animals outside the hospital environment. Other studies have reported the co-occurrence of ESBL and *qnr* genes, suggesting that these genes may be located within the same genetic element, for example, a plasmid (Bajaj et al., 2016; FarajzadehSheikh et al., 2019). The coexistence of ARGs against these two critical antimicrobial classes within the same MGE or strain represents a significant risk, as it can contribute to the survival and persistence of microorganisms under environmental conditions and during clinical treatment, potentially promoting the selection of MDR strains. These findings emphasize the need to monitor the co-occurrence of Plasmid-Mediated Quinolone Resistance (PMQR) and ESBL genes, given their impact on human and animal health and the risk of MDR *E. coli* dissemination in the environment.

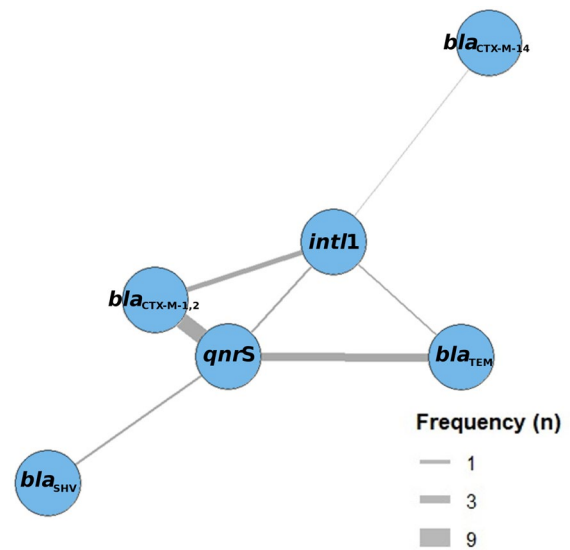


Fig. 7 Network analysis of genes associated with *intI1* and *qnrS* in ESBL-Ec. The thickness of the lines represents the number of times co-occurrence the two genes was detected

Sabino and co-workers (2024) analysed in silico plasmids of *Enterobacteriaceae* from different sources – feces of humans and food-producing animals, sewage and soil. Although plasmids carrying multiple ARGs have been found in all sources, the greatest abundance was found in *Enterobacteriaceae* from chicken meats (Sabino et al., 2024). Despite the majority of AMR solutions being based on management features, like reducing some antimicrobial use, data science approaches are already being implemented in some countries to predict or even monitoring clinical risks (Djordjevic et al., 2024). Whole-genome sequencing (WGS) and knowledge of the bacterial resistome are advancing, but a vast amount of data still needs to be integrated and interpreted, posing a major challenge to the global implementation of these digital technologies, especially in low- and middle-income countries (Majumdar et al., 2026). According to Seymour and McLellan (2025), constant climate change is reshaping the physical, chemical, and biological patterns of natural environments, such as marine ecosystems, and these changes may impact marine microbiomes. The increase in the Earth's temperature (with a consequent increase in water temperature) may also favor the persistence and multiplication of microorganisms with pathogenic potential in these marine environments. In addition, climatic events such

as heavy rainfall and flooding increase the transport of pathogenic microorganisms via sewage and urban and agricultural runoff to natural water sources (Seymour & McLellan, 2025). Although it was not possible to establish a statistical correlation between water temperature and balneability or the number of *E. coli* isolates, this hypothesis reinforces the importance of considering climatic variables in microbiological environmental surveillance studies.

Despite the presence of foodborne pathogens in natural ecosystems has changed over the years, the charge of resistance genes in different ecological niches is of great concern. The resistance gene flow across the food chain, related to the antimicrobial use in poultry and livestock industry, contributes to AMR in humans. The rationale use of antimicrobials, together with genomic surveillance in different ecological niches, could reinforce One Health concept in order to mitigate AMR (Karabasil et al., 2025). All surveillance programs established worldwide aim to collect various types of data from communities and hospitals, and several studies point to the spread of antimicrobial resistance genes (ARGs) between animals and humans through the environment (Doyle et al., 2025; Sanni et al., 2024; Zhang et al., 2025b). The innovations in AMR surveillance may function as a complement to well-known strategies, improving analysis of many isolates across different ecosystems and tracking AMR mechanisms. In the latest sanitation ranking conducted by the Trata Brasil Institute in 2024, Niterói was ranked fifth among the municipalities with the best basic sanitation in Brazil (CBH Baía de Guanabara, 2024). The city has 95.5% coverage of sewage collection and treatment (RJ = 64.9%, Brazil = 55.5%) and 100% coverage of water supply and treatment. These results are a consequence of investment in basic sanitation in the city (Prefeitura Municipal de Niterói, 2024; Prefeitura Municipal de Niterói, 2022; Cidade de Niterói, 2024; Instituto Água e Saneamento, n.d.).

The improvement in sanitation in Niterói over the years has brought great benefits to the city (Neves, 2025). Although the city of Niterói treats a large part of its sewage, untreated sewage from several other cities reaches Guanabara Bay, contaminating the environment. Our study demonstrates the importance of the Tricycle Surveillance Protocol, given that the presence of ESBL-Ec was not directly linked to the detected

pollution. Contamination can stem from various pollution sources; safeguarding environmental, human, and animal health requires not only multisectoral surveillance but also the integration of public policies. To preserve the natural environment of the bay, public policies for treating sewage from surrounding cities must be created. This report pays particular attention to the role of untreated sewage in transmitting potential pathogens and their resistance determinants in urban coastal waters, addressing the presence of these potential AMR Enterobacterales in marine environments through a One Health lens, according to the Tricycle Protocol.

4 Conclusions

Our findings revealed the presence of MDR *E. coli*, including ESBL-producing strains, and carriers of multiple antimicrobial resistance genes in marine ecosystems, regardless of the season or water quality index. These results highlight the importance of the ongoing monitoring of natural marine environments, through integrated multisectoral approaches aligned with One Health framework and the Tricycle Protocol. Future studies, including the investigation of AmpC enzymes, genomic analyses, and plasmid characterization, are essential to advance our understanding of the dissemination of AMR in marine environments. Additionally, our results underscore the importance of implementing public policies for environmental surveillance in conjunction with human and animal health surveillance to mitigate the impact of AMR dissemination in Guanabara Bay.

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Author Contribution M.T.X.S. and J.P.A. collected and processed the samples; M.T.X.S. performed all experiments; J.A.S. assisted in ESB� screening and prepared Fig. 4; A.M.F.C. and J.P.A. were responsible for the conceptualization of the investigation; M.T.X.S. and J.P.A. were responsible for the formal analysis; A.M.F.C., C.B.V., and F.P.G.N. were responsible for project resources and administration; B.F.R.O. assisted in the genotypic analysis; M.T.X.S., B.F.R.O., and J.P.A. drafted the original manuscript; B.F.R.O. and J.P.A. supervised the investigation. All authors reviewed the manuscript.

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Data Availability All data supporting the findings of this study are available within the paper and its Supplementary Information. Water quality classification at the sampling locations was obtained from INEA reports for the years 2022 and 2023, available at <https://www.inea.rj.gov.br/niteroi/> and provided in Supplementary Table 1. All primer sequences used in this study are provided in Supplementary Table 2, along with original reference.

Declarations

Ethical Approval Not applicable.

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